

Greenberg Harper 10

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1 Chain Complexes

Suppose $f : C \rightarrow C', g : C' \rightarrow C''$ are chain maps. Then so is $gf : C \rightarrow C''$.

Proof: Observe that

$$\partial_q'' g_q f_q = g_{q-1} \partial_q' f_q = g_{q-1} f_{q-1} \partial_q,$$

so gf is indeed a chain map.

Suppose $f : C \rightarrow C'$ is a chain map. Define $H(f)$ by $H_q(f) : H_q(C) \rightarrow H_q(C')$ by $H_q(f)(\bar{c}) = \overline{f_q(c)}$. Suppose $c_1, c_2 \in Z_q(C)$ such that $c_1 - c_2 \in B_q(C)$. Then

$$f_q(c_1) - f_q(c_2) = f_q(c_1 - c_2) = f_q \partial_{q+1}(d) = \partial_{q+1}' f_{q+1}(d) \in B_q(C').$$

Suppose $c \in Z_q(C)$. Then

$$\partial_q' f_q(c) = f_{q-1} \partial_q(c) = f_{q-1}(0) = 0,$$

so $f_q(c) \in Z_q(C')$. Combining the two previous results, $H_q(f)$ is well-defined.

Suppose $f : C \rightarrow C', g : C' \rightarrow C''$ are chain maps. Then $H(gf) = H(g)H(f)$.

Proof:

$$H_q(gf)(\bar{c}) = \overline{g_q f_q(c)} = H_q(g)(\overline{f_q(c)}) = H_q(g)H_q(f)(\bar{c}).$$

It follows $H_q(gf) = H_q(g)H_q(f)$.

Suppose $f, g : C \rightarrow C'$ are chain homotopic. Then $H_q(f) = H_q(g)$.

Proof: Let $c \in Z(C)$. We have

$$\begin{aligned} f_q(c) - g_q(c) &= \partial_{q+1}' D_q(c) + D_{q-1} \partial_q(c) \\ &= \partial_{q+1}' D_q(c) \in B_q(C'). \end{aligned}$$

Hence $H_q(f)(c) = \overline{f_q(c)} = \overline{g_q(c)} = H_q(g)(c)$, ie $H_q(f) = H_q(g)$.

Suppose $f_1, f_2 : C \rightarrow C'$ and $g_1, g_2 : C' \rightarrow C''$ are chain homotopies with $f_1 \simeq f_2$ and $g_1 \simeq g_2$. Then $g_1 f_1 \simeq g_2 f_2$.

Let $f_1 \simeq f_2, g_1 \simeq g_2$ by the diagonal maps D_f, D_g . Write

$$\begin{aligned} g_1 f_1 - g_2 f_2 &= g_1 f_1 - g_2 f_1 + g_2 f_1 - g_2 f_2 \\ &= (g_1 - g_2) f_1 + g_2 (f_1 - f_2) \\ &= (\partial'' D_g + D_g \partial') f_1 + g_2 (\partial' D_f + D_f \partial) \\ &= \partial'' D_g + D_g f_1 \partial + \partial'' g_2 D_f + g_2 D_f \partial \\ &= \partial'' (D_g + g_2 D_f) + (D_g f_1 + g_2 D_f) \partial. \end{aligned}$$

Chain homotopy is an equivalence relation.

(1). Sps $f : C \rightarrow C'$. Then $f \simeq f$. Proof: Let $D = 0$. Then

$$f - f = 0 = \partial' D + D \partial = 0.$$

(2). Sps $f, g : C \rightarrow C'$ st $f \simeq g$ via diagonal maps D . Then

$$g - f = -(f - g) = -(\partial' D + D \partial) = \partial' (-D) + (-D) \partial.$$

(3). Sps $f, g, h : C \rightarrow C'$ st $f \simeq g$ via D_a and $g \simeq h$ via D_b . Then

$$\begin{aligned} f - h &= (f - g) + (g - h) = (\partial' D_a + D_a \partial) + (\partial' D_b + D_b \partial) \\ &= \partial' (D_a + D_b) + (D_a + D_b) \partial. \end{aligned}$$

Sps $Q \in X$ is a point. Define $c : X \rightarrow X$ to be the constant map at Q . Find $S_q(c) : S_q(X) \rightarrow S_q(X)$. Define $c' : S_q(X) \rightarrow S_q(X)$ by $c' = 0$ for $q > 0$ and $c'(\sigma) = \sigma_0$ for $q = 0$. Show that $S_q(c) \simeq c'$.

Denote by $\sigma_q : \Delta_q \rightarrow X$ be the constant map at Q . Note that σ_0 coincides with the above definition of σ_0 . Observe that $S_q(c)(\sigma) = \sigma_q$ for all simplicial q -simplices σ . Define $D_q : S_q(X) \rightarrow S_{q+1}(X)$ by $D_q(\sigma) = \sigma_{q+1}$. I claim that $S(c) \simeq c'$ via D .

Note for $q = 0$, we have

$$S_0(c)(\sigma) - c'(\sigma) = \sigma_0 - \sigma_0 = 0$$

and

$$\partial_1 D_0(\sigma) = \partial_1 \sigma_1 = 0$$

(since 1 is odd).

Now consider $q > 0$. We have

$$S_q(c)(\sigma) - c'(\sigma) = \sigma_q$$

Suppose q is even. Write $\partial_q(\sigma) = \sum_{i=0}^q \sigma F_q^i$, so $\partial_q(\sigma)$ is the alternating sum of $q+1$ simplicial $q-1$ -simplices. $D_{q-1}\partial_q(\sigma) = \sigma_q$. Note that

$$\partial_{q+1} D_q(\sigma) = \partial_{q+1} \sigma_{q+1} = 0,$$

since $q+1$ is odd. Hence, $S_q(c) - c' = \partial_{q+1} D_q + D_{q-1} \partial_q$ for even q .

Suppose q is odd. We compute $D_{q-1}\partial_q(\sigma) = 0$ and $\partial_{q+1} D_q(\sigma) = \partial_{q+1} \sigma_{q+1} = \sigma_q$ (since $q+1$ is even). Hence, $S_q(c) - c' = \partial_{q+1} D_q + D_{q-1} \partial_q$ for all $q \geq 0$.

2 Relation Between π_1 and H_1

Let $\chi : \pi_1(X, x_0) \rightarrow H_1(X)$ be defined by $\chi([\gamma]) = \bar{\gamma}$.

Claim: χ is well-defined. Suppose $\gamma \simeq_p \gamma'$ by the homotopy F_1 . Define $\sigma_1 : \Delta_1 \rightarrow X$ by $\sigma_1(E_0(1-t) + E_1 t) = \gamma(t)$. Define σ'_1 similarly. I claim $\sigma_1 \simeq \sigma'_1$.

Define $F_2 : \Delta_1 \times I \rightarrow X$ by $F_2((1-s)E_0 + sE_1, t) = F_1(s, t)$. For $t = 0$, $F_2((1-s)E_0 + sE_1, 0) = \gamma(s)$. For $t = 1$, $F_2((1-s)E_0 + sE_1, 1) = \gamma'(s)$, so F_2 is a homotopy from σ_1 to σ'_1 .

Then $H_1(\sigma_1) = H_1(\sigma'_1)$, so $\overline{\sigma_1 \delta_1} = \overline{\sigma'_1 \delta_1}$, ie $\overline{\sigma_1}$, or $\overline{\sigma_1} = \overline{\sigma'_1}$, so χ is well-defined.

3 The Exact Homology Sequence

Suppose

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{j} C \longrightarrow 0$$

is short exact such that there exists $k : B \rightarrow A$ such that $ki = \text{id}_A$. Define $l : C \rightarrow B$ by $l(c) = b - ik(b)$ for some $b \in B$ such that $j(b) = c$ (since j is epi).

I claim that $l(c)$ is independent of b . Suppose $j(b_1) = j(b_2) = c$. Then $j(b_1 - b_2) = 0$. Since the sequence is exact, let $b_1 - b_2 = i(a)$ for some $a \in A$.

Then $ik(b_1 - b_2) = iki(a) = i(a) = b_1 - b_2$, so $b_1 - ik(b_1) = b_2 - ik(b_2)$.

Suppose $c_1, c_2 \in C$, $j(b_1) = c_1$, $j(b_2) = c_2$. Note that $j(b_1 + b_2) = c_1 + c_2$, so

$$\begin{aligned} l(c_1 + c_2) &= b_1 + b_2 - ik(b_1 + b_2) = (b_1 - ik(b_1)) + (b_2 - ik(b_2)) \\ &= l(c_1) + l(c_2). \end{aligned}$$

If $c \in C$, $r \in R$, $b \in B$ such that $j(b) = c$, then

$$l(rc) = rb - ik(rb) = r(b - ik(b)) = rl(c),$$

so l is a module homomorphism.

Further,

$$jl(c) = j(b - ik(b)) = j(b) = c,$$

so the other condition is satisfied.

Conversely, suppose that $jl = \text{id}_C$. Define $k : B \rightarrow A$ by $k(b) = i^{-1}(b - lj(b))$. Note that $j(b - lj(b)) = j(b) - j(b) = 0$, so $i^{-1}(b - lj(b)) \neq \emptyset$. Since i is injective, k is well-defined. k is a module homomorphism and $ki(a) = i^{-1}(i(a) - lji(a)) = a$, so $ki = \text{id}_A$.

Suppose $k(b) = 0$. Then $l(j(b)) = b - ik(b) = b$, so $b \in \text{im } l$. Suppose $b = l(c)$ for some $c \in C$. Let $c = j(b')$ for some $b' \in B$. Then $b = b' - ik(b')$, so $k(b) = k(b') - k(b') = 0$.

Sps $g \in G$. $j_1(g) = k_2(g') = j_1i_2(g') \rightarrow j_1(g - i_2(g')) = 0 \rightarrow g - i_2(g') = i_1(g'')$, so $g = \nabla'(i_1 \oplus i_2)(g'', g')$.

Claim $(j_1 \oplus j_2) \circ \Delta$ is isom $G \rightarrow G'_1 \oplus G'_2$.

Sps $(g'_1, g'_2) \in G'_1 \oplus G'_2$. Try $(k_1^{-1}(g'_1), k_2^{-1}(g'_2))$.

4 Exact Sequences

Let $0 \rightarrow \mathcal{C} \xrightarrow{f} \mathcal{D} \xrightarrow{g} \mathcal{E} \rightarrow 0$ be a short exact sequence of chain complexes. Then there exists a long exact homology sequence:

$$\cdots \rightarrow H_n(C) \rightarrow H_n(D) \rightarrow H_n(E) \rightarrow H_{n-1}(C) \rightarrow H_{n-1}(D) \rightarrow H_{n-1}(E) \rightarrow \cdots$$

Proof: Let the maps $H_n(C) \xrightarrow{H_n(f)} H_n(D) \xrightarrow{H_n(g)} H_n(E)$ be as usual. Define the connecting homomorphisms $d_n : H_n(E) \rightarrow H_{n-1}(C)$ as follows:

Let $z \in Z_n(E)$. Since g_n is surjective, there exists $y \in D_n$ such that $g_n(y) = z$. Note that $\partial g_n(y) = g_{n-1}\partial y = 0 \rightarrow \partial y = f_{n-1}(x)$ for some $x \in C_{n-1}$ (by exactness). Now $0 = \partial\partial y = \partial f_{n-1}(x) = f_{n-2}\partial x$. By injectivity of f_n , we have $\partial x = 0$, so $x \in Z_{n-1}(C)$. Define $d_n(\bar{z}) = \bar{x}$. I claim that d_n is well-defined.

Suppose $z_1 - z_2 \in B_n(E)$, so $z_1 - z_2 = \partial z$ for some $z \in E_{n+1}$. By surjectivity of g_{n+1} , let $y \in D_{n+1}$ such that $g_{n+1}(y) = z$. Then $g_n(y_1 - y_2) = \partial g_{n+1}(y) = g_n\partial y \rightarrow y_1 - y_2 - \partial y = f_n(x)$ for some $x \in C_n$ by exactness. Taking boundaries, we have $\partial y_1 - \partial y_2 = \partial f_n(x) = f_{n-1}\partial x$. $LHS = f_{n-1}(x_1 - x_2)$, so $x_1 - x_2 = \partial x$, so d is indeed well-defined.

It should be straightforward to check that d_n is a module homomorphism.

(1). Exactness at node $H_n(D)$. Suppose $H_n(g)(\bar{y}) = 0$ for some $y \in Z_n(D)$, so $g_n(y) = 0$, so $g_n(y) \in B_n(E)$. Write $g_n(y) = \partial z$ for some $z \in E_{n+1}$. By surjectivity of g_n , let $z = g_{n+1}(y')$ for some $y' \in D_{n+1}$. Then $\partial z = \partial g_{n+1}(y') = g_n\partial y'$, so $g_n(y - \partial y') = 0$. By exactness, let $y - \partial y' = f_n(x)$ for some $x \in C_n$. Note that $f_{n-1}\partial x = \partial f_n(x) = \partial y = 0$, so by injectivity, $\partial x = 0$. Hence, $x \in Z_n(C)$. In other words, $H_n(f)(\bar{x}) = \bar{y}$.

(2). Exactness at node $H_n(E)$. Suppose $d_n(\bar{z}) = 0$ for some $z \in Z_n(E)$. Let $g_n(y) = z$ by surjectivity of g_n . We have $\partial y = f_{n-1}(x)$ for some $x \in B_{n-1}(C)$, ie $x = \partial x'$ for some $x' \in C_n$.

We have $\partial y = f_{n-1}(x) = f_{n-1}(\partial x') = \partial f_n(x')$, so $\partial(y - f_n(x')) = 0$, so $y - f_n(x') \in Z_n(D)$. Further,

$$g_n(y - f_n(x')) = g_n(y) = x,$$

so $H_n(g)(\overline{y - f_n(x)}) = \bar{x}$.

(3). Exactness at node $H_n(C)$. Suppose $x \in Z_n(C)$ such that $H_n(f)(\bar{x}) = 0$, ie $f_n(x) = \partial y$ for some $y \in D_{n+1}$. Let $z = g_{n+1}(y)$. Note that $\partial z = \partial g_{n+1}(y) = g_n\partial y = g_nf_n(x) = 0$, so $z \in Z_{n+1}(E)$. Hence, we have $d_{n+1}(\bar{z}) = \bar{x}$, and we're done.

Suppose $C \supset B \supset A$. Consider $0 \rightarrow S$